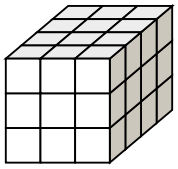
Preview: set-03/by_level/Grade_03-04/y2003_q09 (v4)

Inside of a box of cubes: every length drops by 2; outside = total − inside

Part A: Each question is worth 3 points

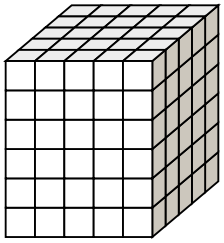
1. Sara built this brick (3 cubes long, 4 cubes deep and 3 cubes tall) from red and blue cubes of the same size. Every cube you can see from the outside (also from below and behind) is red, and all the cubes inside are blue.



How many blue cubes did Sara use?

A. 1 ***B.*** 2 ***C.*** 4 ***D.*** 12 ***E.*** 34

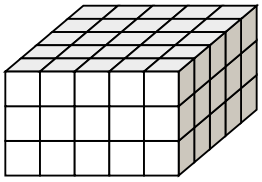
2. Chris built this brick (5 cubes long, 5 cubes deep and 6 cubes tall) from red and blue cubes of the same size. Every cube you can see from the outside (also from below and behind) is red, and all the cubes inside are blue.



How many blue cubes did Chris use?

A. 9 ***B.*** 12 ***C.*** 30 ***D.*** 34 ***E.*** 36

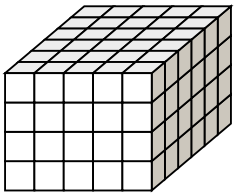
3. Sara built this brick (5 cubes long, 5 cubes deep and 3 cubes tall) from red and blue cubes of the same size. Every cube you can see from the outside (also from below and behind) is red. All the cubes inside are blue.



How many red cubes did Sara use?

A. 9 ***B.*** 43 ***C.*** 55 ***D.*** 58 ***E.*** 66

4. Leo built this brick (5 cubes long, 6 cubes deep and 4 cubes tall) from red and blue cubes of the same size. Every cube you can see from the outside (also from below and behind) is red. All the cubes inside are blue.

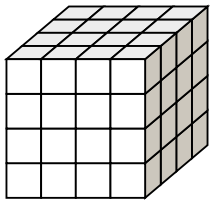


How many red cubes did Leo use?

A. 24 ***B.*** 74 ***C.*** 88 ***D.*** 96 ***E.*** 120

Part B: Each question is worth 4 points

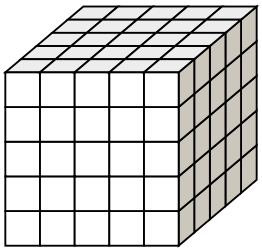
5. Nina glued 64 small cubes into a big 4 × 4 × 4 cube (like the picture) and painted the whole outside of it, also the bottom. Then the big cube falls apart.



How many small cubes have no paint at all?

A. 4 ***B.*** 5 ***C.*** 6 ***D.*** 7 ***E.*** 8

6. Nina glued 125 small cubes into a big 5 × 5 × 5 cube (like the picture) and painted the whole outside of it, also the bottom. Then the big cube falls apart.



How many small cubes have paint on exactly three faces?

A. 36 ***B.*** 27 ***C.*** 18 ***D.*** 12 ***E.*** 8

7. Sara builds a brick 4 cubes long and 3 cubes tall from red and blue cubes. The cubes you can see from the outside (also from below and behind) are red and all the cubes inside are blue. Sara used exactly 6 blue cubes.

How many cubes deep is the brick?

A. 2 ***B.*** 3 ***C.*** 4 ***D.*** 5 ***E.*** 7

8. Sara builds a brick 5 cubes long and 5 cubes tall from red and blue cubes. The cubes you can see from the outside (also from below and behind) are red and all the cubes inside are blue. Sara used exactly 18 blue cubes.

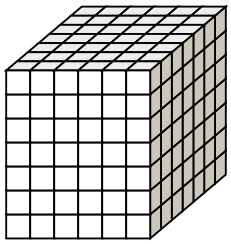
How many cubes deep is the brick?

A. 1 ***B.*** 2 ***C.*** 3 ***D.*** 4 ***E.*** 6

Part C: Each question is worth 5 points

9. Tom builds a brick 6 cubes long, 7 cubes deep and 7 cubes tall from red and blue cubes.

- All the cubes you can see from the outside are red.
- The cubes just below them are **also** red, so on every side (top, bottom, front, back, left and right) there are two layers of red cubes.
- All the other cubes, in the middle, are blue.

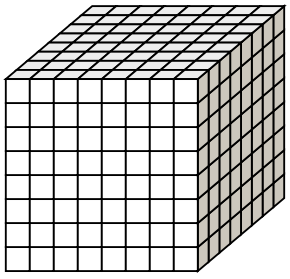


How many blue cubes are there?

A. 100 ***B.*** 18 ***C.*** 15 ***D.*** 14 ***E.*** 6

10. Leo builds a brick 8 cubes long, 8 cubes deep and 8 cubes tall from red and blue cubes.

- All the cubes you can see from the outside are red.
- The cubes just below them are **also** red, so on every side (top, bottom, front, back, left and right) there are two layers of red cubes.
- All the other cubes, in the middle, are blue.



How many blue cubes are there?

A. 16 ***B.*** 55 ***C.*** 58 ***D.*** 61 ***E.*** 64

Answer Key

1. **B**	2. **E**	3. **E**	4. **D**	5. **E**	6. **E**
7. **D**	8. **D**	9. **B**	10. **E**		

Solutions

1. B (2) — Take away one layer on every side: the inside is 1 × 2 × 1 = 2. [set-03/by_level/Grade_03-04/2003 Q9 · blue_inside]

2. E (36) — Take away one layer on every side: the inside is 3 × 3 × 4 = 36. [set-03/by_level/Grade_03-04/2003 Q9 · blue_inside]

3. E (66) — All cubes: 5 × 5 × 3 = 75. Blue inside: 3 × 3 × 1 = 9. Red: 75 – 9 = 66. [set-03/by_level/Grade_03-04/2003 Q9 · red_outside]

4. D (96) — All cubes: 5 × 6 × 4 = 120. Blue inside: 3 × 4 × 2 = 24. Red: 120 – 24 = 96. [set-03/by_level/Grade_03-04/2003 Q9 · red_outside]

5. E (8) — They are the cubes inside: 2 × 2 × 2 = 8. [set-03/by_level/Grade_03-04/2003 Q9 · painted]

6. E (8) — They are the corners, and a cube always has 8 corners. [set-03/by_level/Grade_03-04/2003 Q9 · painted]

7. D (5) — The blue inside is 2 long and 1 tall, so one layer of it has 2 × 1 = 2 cubes. 6 ÷ 2 = 3 layers of blue, plus the red front and back: 5. [set-03/by_level/Grade_03-04/2003 Q9 · how_deep]

8. D (4) — The blue inside is 3 long and 3 tall, so one layer of it has 3 × 3 = 9 cubes. 18 ÷ 9 = 2 layers of blue, plus the red front and back: 4. [set-03/by_level/Grade_03-04/2003 Q9 · how_deep]

9. B (18) — Each length loses two red cubes at each end, so it drops by 4: 2 × 3 × 3 = 18. [set-03/by_level/Grade_03-04/2003 Q9 · thick_wall]

10. E (64) — Each length loses two red cubes at each end, so it drops by 4: 4 × 4 × 4 = 64. [set-03/by_level/Grade_03-04/2003 Q9 · thick_wall]